# Early warning for students at risk of failing

Mohamed Reda Zarrouk and Georgy Salameh · PGE 2 final project, Machine Learning × Advanced Python · November 2026 · Code: `student_risk` package, analysis in `notebooks/analysis.ipynb`

## 1. Problem

**User and decision.** The tool is meant for a secondary school's teaching team at the **start of the school year**. For each student it estimates the probability of failing the year, and flags the students to whom the school should **offer tutoring** before any grade of the year is known.

**Target and task.** `fail = 1` if the final grade G3 is below 10 out of 20, otherwise 0. This is a binary classification task in which "fail" is the positive and minority class (15.4 % of students).

**Cost of each mistake.** A **false negative** (a failing student who is not flagged) gets no help and may fail the year: this is the costly error. A **false positive** (a passing student who is flagged) costs some tutoring time, and extra help rarely harms a student. We therefore put **recall first**: we require a recall of at least 0.80 (set in `config.yaml`) and, among the thresholds that reach it, we choose the one with the best precision, so that tutors are not flooded with false alarms.

## 2. Data

We use the UCI *Student Performance* dataset (id 320, CC BY 4.0; Cortez & Silva, 2008), loaded with `ucimlrepo` by our `DatasetLoader` class. It has **649 students** and 33 columns: 30 features plus the three period grades G1, G2 and G3.

- **Types and quality:** 13 numeric and 17 text (categorical) features; no missing value and no duplicate row. The imputers in the pipeline are kept as a safeguard for new cases, for which `predict` accepts empty fields.
- **Class balance:** 100 fails and 549 passes. A model that always answers "pass" is 85 % accurate and useless, so accuracy is never used.
- **Special code G3 = 0:** 15 students have a final grade of exactly 0, and 8 of them had a non-zero G2. A zero after a non-zero period grade suggests that the student may not have sat the final exam rather than answered everything wrong. We **keep them as fails**: for the school, a student who ends the year with 0 did not pass, and missing the exam is itself a sign of risk. Section 5 checks that this choice does not distort the model.
- **Leakage:** G1 and G2 have correlations of 0.83 and 0.92 with G3. A model using them would mostly copy G2, and above all they are **not known at the start of the year**, when the warning is needed. Both are dropped, and G3 is dropped once the target is built. A test checks that none of the three is among the features.
- **Split:** one stratified train/test split (20 % test, seed 42): **519 training students (80 fails)** and **130 test students (20 fails)**, with no student in both sets. The test set was then left untouched until the final evaluation.

## 3. Method

**Pipeline.** All preparation is inside one scikit-learn `Pipeline` whose first step is a `ColumnTransformer`: numeric columns get median imputation then standardisation; text columns get most-frequent imputation then one-hot encoding (56 columns, unknown categories ignored). Because the whole pipeline is refitted on each training fold, the scaler and the encoder never see the validation fold, so no information leaks into the cross-validation scores.

**Protocol.** Every choice was made on the training set with the **same 5-fold `StratifiedKFold`** (seed 42) for every model. We started from a `DummyClassifier` (always "pass") and an untuned logistic regression, then compared four models, each tuned with `GridSearchCV` (grids in `config.yaml`):

- logistic regression with an **L2** penalty and with an **L1** penalty: $C \in \{0.001, 0.01, 0.1, 1, 10\}$, class weight none or balanced;

- **linear SVM** (`LinearSVC` inside `CalibratedClassifierCV` to obtain probabilities): same C and class-weight grid;
- **k-NN**: k ∈ {5, 11, 21, 31, 41, 51}, uniform or distance weights.

**Metric.** Models are refitted and compared on **average precision** (AP, the area under the precision-recall curve). AP focuses on the rare positive class and does not depend on a threshold, which is chosen separately. ROC-AUC, recall, precision and F1 are reported as well.

**Selection rule.** We take the model with the best mean CV AP, but if a simpler model is within one standard deviation of it, we prefer the simpler one. The simplicity order is L2 logistic regression < L1 logistic regression < linear SVM < k-NN.

**Threshold.** `Trainer` computes **out-of-fold** probabilities for the 519 training students (each student is predicted by a model that did not see them, on the same five folds) and chooses the threshold with recall ≥ 0.80 and the best precision.

**Single test.** The chosen pipeline, refitted on the whole training set, and the chosen threshold are scored **once** on the 130 test students, with a confusion matrix and a 95 % Wilson interval for the test recall. Nothing was changed afterwards.

## 4. Results

### Baseline and model comparison (5-fold CV, training set)

| Model | Best hyperparameters | AP (std) | ROC-AUC | Train AP | Recall @0.5 |
|---|---|---|---|---|---|
| Dummy (always "pass") | – | 0.154 | 0.500 | – | 0.00 |
| Logistic regression, untuned | defaults | 0.457 | 0.779 | – | 0.30 |
| **Logistic regression L2 (chosen)** | **C = 0.01, no class weight** | **0.500 (0.143)** | **0.803** | **0.560** | **0.05** |
| Logistic regression L1 | C = 0.1, no class weight | 0.538 (0.158) | 0.828 | 0.564 | 0.24 |
| Linear SVM (calibrated) | C = 0.001, balanced | 0.500 (0.147) | 0.803 | 0.552 | 0.23 |
| k-NN | k = 31, distance weights | 0.510 (0.179) | 0.768 | 1.000 | 0.05 |

Table 1. Source: `reports/baseline.csv` and `reports/model_comparison.csv`, written by `student-risk train`. AP of the dummy equals the fail rate. Recall at the default threshold 0.5 is shown only to motivate the threshold choice.

Every tuned model ranks students far better than the dummy. L1 logistic regression has the best mean AP (0.538), but its standard deviation is 0.158 and **all four models are within one standard deviation of it**: with 80 fails in the training set, the folds cannot separate them. The rule therefore picks the simplest model, **L2 logistic regression with C = 0.01**. k-NN clearly overfits (train AP 1.000 against 0.510 in CV). Because the hyperparameters and the model were chosen on the same folds, the CV score of the chosen model is slightly optimistic.

### Decision threshold (out-of-fold, training set)

| Rule | Threshold | Recall | Precision | Flagged |
|---|---|---|---|---|
| Default | 0.500 | 0.050 | 0.571 | 1.3 % |
| Recall ≥ 0.70 | 0.171 | 0.725 | 0.389 | 28.7 % |
| **Recall ≥ 0.80 (chosen)** | **0.140** | **0.800** | **0.312** | **39.5 %** |
| Recall ≥ 0.90 | 0.116 | 0.900 | 0.248 | 55.9 % |

Table 2. Source: `reports/threshold_report.csv` and `reports/threshold_tradeoff.csv`.

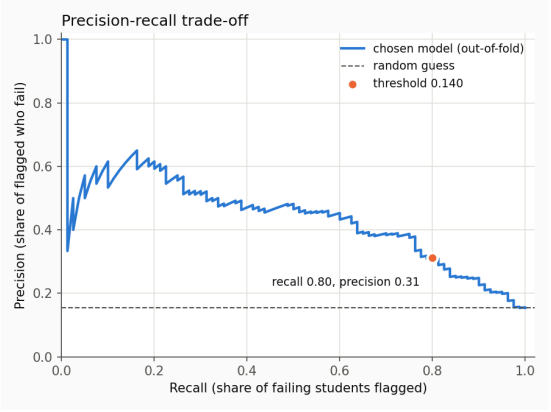



Figure 1. Out-of-fold precision-recall curve of the chosen model; the point marks the chosen threshold.

At 0.5 the model catches only 4 of the 80 failing students, because the strong regularisation keeps its probabilities low. At the chosen threshold of **0.140** it catches 64 of 80 (recall 0.80) with precision 0.31, which means flagging 39.5 % of the students. Table 2 shows the price of recall, so that a school can pick the row that matches its tutoring capacity.

### Single test evaluation (130 students, used once)

| Metric | CV / out-of-fold (train) | Test |
|---|---|---|
| ROC-AUC | 0.803 | 0.778 |
| Average precision | 0.500 | 0.346 |
| Recall at 0.140 | 0.800 | **0.750** (15 / 20) |
| Precision at 0.140 | 0.312 | 0.300 |
| F1 at 0.140 | 0.449 | 0.429 |
| Share of students flagged | 39.5 % | 38.5 % |

Table 3. Confusion matrix on the test set: TN 75, FP 35, FN 5, TP 15.

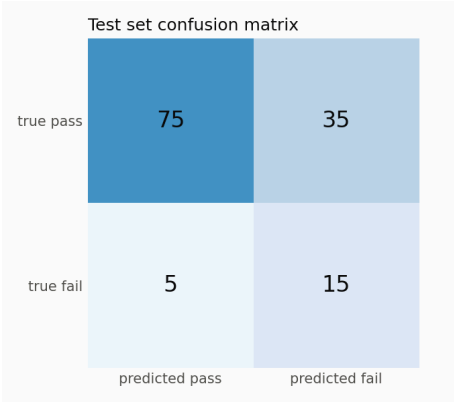



Figure 2. Confusion matrix on the test set at threshold 0.140.

The model catches **15 of the 20** failing test students and misses 5, with 35 false alarms. The recall of 0.75 is just below the 0.80 goal, but with only 20 failing students its **95 % Wilson interval is [0.53, 0.89]**: the test set cannot tell a true recall of 0.75 from one of 0.80. Precision and the flagged share match the out-of-fold estimates closely. The drop in AP (0.500 to 0.346) is consistent with the optimism of model selection and with the small test set.

## 5. Interpretation

**Reading the coefficients.** A coefficient of the logistic regression is a change in the **log-odds of failing**. Numeric features are standardised, so a coefficient is the effect of +1 training standard deviation, and $e^{\text{coefficient}}$ is the factor applied to the odds. Text features are one-hot encoded with every category kept; with the L2 penalty, the two columns of a yes/no feature get opposite coefficients, and what matters is their **difference**. Coefficients are small overall because C = 0.01 is a strong penalty. They describe **associations, not causes**.

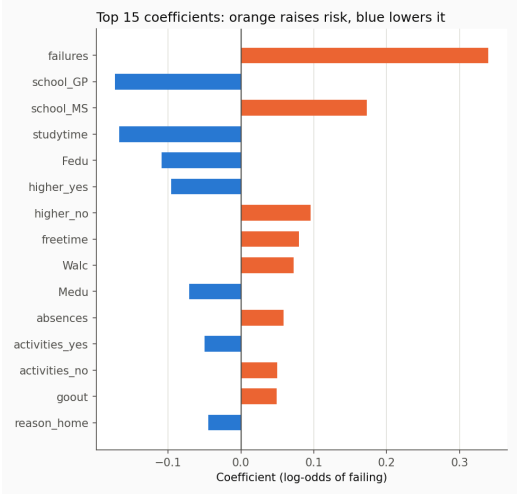



Figure 3. The 15 largest coefficients of the chosen model (positive = higher risk of failing).

- **Past failures (+0.340):** the strongest signal. One standard deviation more (about 0.6 past failures) multiplies the odds of failing by 1.40.
- **School (MS +0.173, GP −0.173):** attending MS rather than GP adds 0.346 to the log-odds (odds × 1.41). In the training set, 31 % of MS students fail against 7 % at GP.
- **Study time (−0.167):** more weekly study time goes with a lower risk.
- **Parents' education (Fedu −0.109, Medu −0.071):** more educated parents go with a lower risk.
- **Wants higher education (no +0.096, yes −0.096):** students who do not plan to continue their studies are at higher risk.
- **Free time (+0.080), weekend alcohol (+0.073), absences (+0.058), going out (+0.049):** each goes with a slightly higher risk.

### Checks on the training set (reported only; nothing was changed)

**Sensitivity to the G3 = 0 students.** With the same model and hyperparameters and the same 5-fold CV, removing the 11 training students with G3 = 0 leaves ROC-AUC almost unchanged (0.803 → 0.805). AP goes from 0.500 to 0.464, but AP depends on the fail rate, which also changes (15.4 % → 13.6 %), and the difference is well within one standard deviation. Nine of the 11 G3 = 0 students are flagged out-of-fold, so they are not hard cases. Our decision to keep them stands.

**Fairness.** We applied the chosen threshold to the out-of-fold probabilities separately for each group:

| Group | Students | Fails | Recall [95 % Wilson] | Precision | Flagged |
|---|---|---|---|---|---|
| Female | 303 | 40 | 0.75 [0.60, 0.86] | 0.28 | 35.0 % |
| Male | 216 | 40 | 0.85 [0.71, 0.93] | 0.34 | 45.8 % |
| School GP | 339 | 25 | 0.68 [0.48, 0.83] | 0.22 | 23.3 % |
| School MS | 180 | 55 | 0.85 [0.74, 0.92] | 0.37 | 70.0 % |

Table 4. Out-of-fold results on the 519 training students, threshold 0.140.

Because the model uses the school as a strong signal, **70 % of MS students are flagged**, whereas a failing GP student is more likely to be missed. Boys are flagged more often than girls. Each recall is based on only 25 to 55 failing students, so the intervals are wide and overlap: these gaps are **warnings to monitor, not proven biases**.

## 6. Limits

**1. Small data and optimistic selection.** With 519 training students and 80 fails, CV scores vary strongly from fold to fold (AP standard deviation ≈ 0.14) and the four models are statistically tied. Choosing the model on the same folds makes its CV score optimistic (AP 0.500 in CV, 0.346 on test), and the test set has only 20 fails, so the

test recall is uncertain (interval [0.53, 0.89]). A larger dataset, or repeated or nested cross-validation, would give more reliable estimates.

**2. Limited signal without grades.** Once G1 and G2 are excluded, the profile features carry only moderate information: reaching a recall of 0.80 means flagging about 40 % of all students, and only about one flagged student in three actually fails. The model is linear and cannot capture interactions between features unless they are added by hand. It was trained on two schools over one period, so it may not transfer to other schools, countries or years without being retrained and checked again.

## 7. Risks of using it on real students

The errors fall on real people. About one failing student in four is missed (5 of 20 in the test set), so "not flagged" must never be read as "safe"; and about two flagged students in three would have passed, so a flag must lead only to an **offer of support**, never to a sanction, a lower expectation or a label in the student's file. The model also relies on the school attended, past failures and parents' education, which reflect social background as much as individual effort; the group rates in Table 4 already differ, so the tool could reinforce existing inequalities unless outcomes are checked per group regularly and a teacher keeps the final decision. Finally, the inputs are sensitive personal data (family situation, parents' jobs and education, alcohol use, health, romantic relationships): collecting them needs a clear legal basis, informed students and families, access limited to the staff who act on the flags, and a strict retention period. Features that are not needed, or that are too intrusive, should be removed even at some cost in accuracy.

## Reproducibility and references

All results come from the `student_risk` package: `pip install -e .`, then `student-risk train` (writes `reports/*.csv` and the model) and `student-risk evaluate` (test metrics). The notebook calls the same package and checks that it recovers exactly the saved threshold and tables. Seed 42 everywhere; 19 pytest tests pass.

P. Cortez, *Student Performance* [Dataset], UCI Machine Learning Repository, 2008. https://doi.org/10.24432/C5TG7T. Licence CC BY 4.0.

P. Cortez and A. M. G. Silva, "Using Data Mining to Predict Secondary School Student Performance", *Proceedings of the 5th Annual Future Business Technology Conference (FUBUTEC 2008)*, 2008.

E. B. Wilson, "Probable inference, the law of succession, and statistical inference", *Journal of the American Statistical Association*, 22(158), 1927.

Generative AI use is declared in `AI_USAGE.md`.